Predictive Modeling of NFL Player Fantasy Points Using Deep Learning and Traditional ML

Evan Ellington
eellingt@stanford.edu
Department of Computer Science, Stanford University
Stanford Computer Science

Project Overview

Fantasy football is a game where participants draft real NFL players to create virtual teams that compete based on those players' weekly on-field performances. Success depends heavily on drafting the right players, since weekly scores come directly from each player's real-world performance. This project evaluates six predictive models designed to estimate players' season-long fantasy point totals.

Commonly used ML models for a task of this type are: a Support Vector Regression Model (SVR), a Random Forest Regression Model (RFR), a Gradient Boosting Regression Model (GBR), and an XGBoost Model (XGB). I set out to make a model for drafting a team that used a neural network and compared it to the above methods.

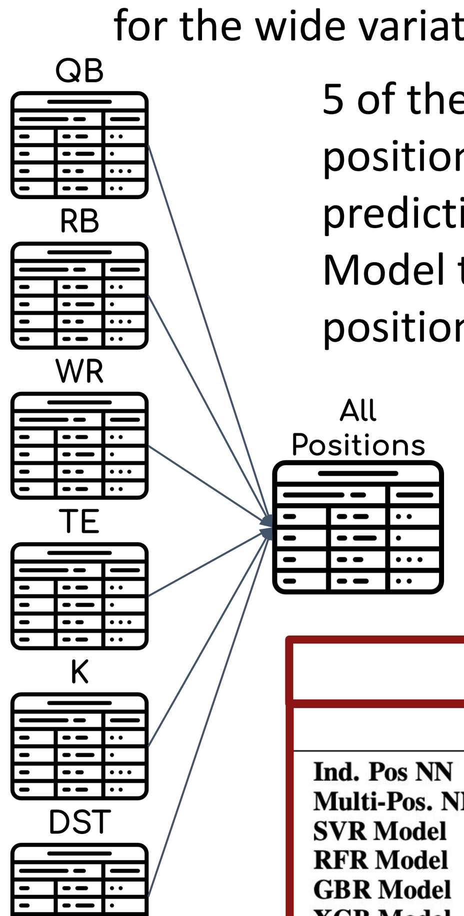
To test how well each model would perform, I simulated a typical snake draft where the model is drafting for one team, and the rest of the opponents are drafting the best available player based on average draft position (ADP).

Round

Team	1st	2nd	3rd	4th	5th	6th	7th	8th	9th	10th	11th	12th	13th	14th	15th
Model	1	24	25	48	49	72	73	96	97	120	121	144	145	168	169
Opponent	2	23	26	47	50	71	74	95	98	119	122	143	146	167	170
Opponent	3	22	27	46	51	70	75	94	99	118	123	142	147	166	171
Opponent	4	21	28	45	52	69	76	93	100	117	124	141	148	165	172
Opponent	5	20	29	44	53	68	77	92	101	116	125	140	149	164	173
Opponent	6	19	30	43	54	67	78	91	102	115	126	139	150	163	174
Opponent	7	18	31	42	55	66	79	90	103	114	127	138	151	162	175
Opponent	8	17	32	41	56	65	80	89	104	113	128	137	152	161	176
Opponent	9	16	33	40	57	64	81	88	105	112	129	136	153	160	177
Opponent	10	15	34	39	58	63	82	87	106	111	130	135	154	159	178
Opponent	11	14	35	38	59	62	83	86	107	110	131	134	155	158	179
Opponent	12	13	36	37	60	61	84	85	108	109	132	133	156	157	180

Datasets & Metrics

My dataset included position-specific statistics from the 2023, 2024, and partial 2025 NFL seasons, compiled from multiple online sources. Each position had its own data frame, and I trained the models using an 80–85% train split with feature scaling to account for the wide variation in raw positional stats.



5 of the 6 models took in the position-separated data sets and made predictions, while the Multi-Positional NN Model took in one data set consisting of all positions.

To evaluate the performance of each model, I computed the RSME value and the R² for the predictions.

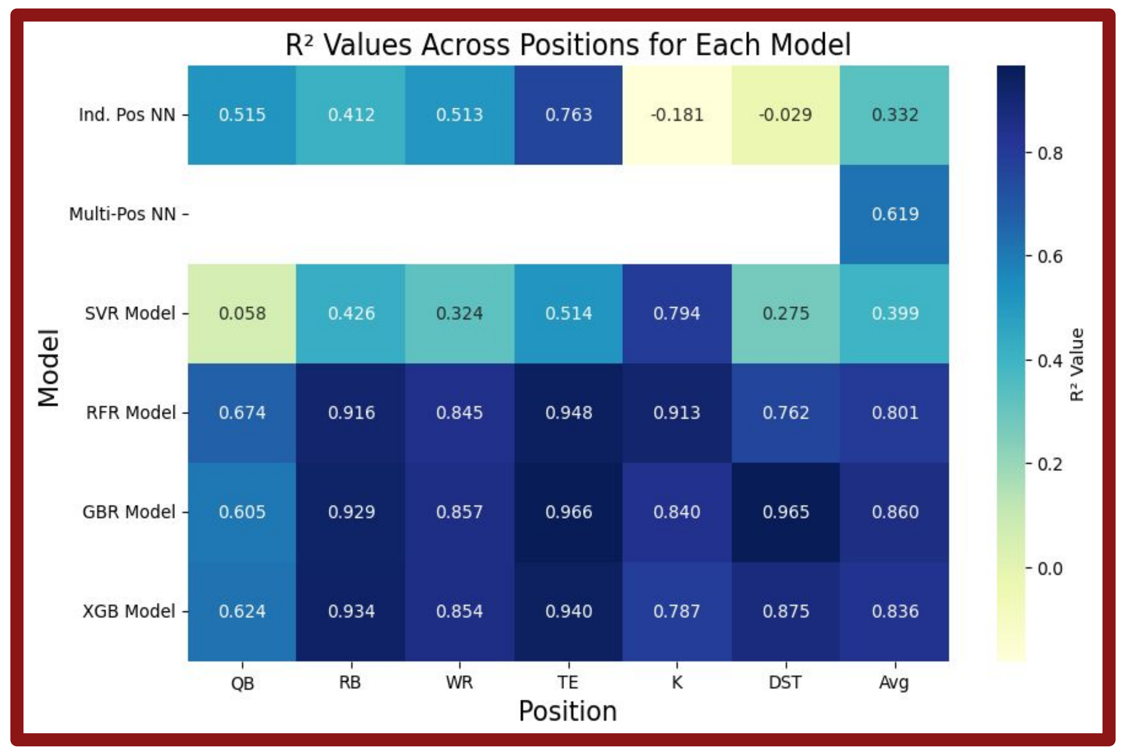
RSME Value for Each Model and Position

	QB	RB	WR	TE	K	DST	Avg RMSE
Ind. Pos NN	121.143	73.907	42.571	32.498	17.099	35.039	53.709
Multi-Pos. NN	-	-	-	-	-	-	65.162
SVR Model	162.143	79.782	63.900	44.444	12.194	21.525	63.998
RFR Model	95.317	30.523	30.567	14.603	7.927	12.332	31.878
GBR Model	104.989	28.079	29.334	11.833	10.741	4.744	31.620
XGB Model	102.384	27.104	29.712	15.589	12.409	8.920	32.686

Methods & Experiments

I trained six predictive models using position-specific NFL data in total. The Individual Positional NN had separate neural networks for quarterbacks (QB), running backs (RB), wide receivers (WR), tight ends (TE) and for kickers (K), defense (DST) due to differences in dataset size and feature structure. The QB/RB/WR/TE neural networks were fully connected neural networks with three hidden layers containing 16, 8, 4 nodes , a dropout layer with rate 0.3 and a single output layer for predicting fantasy points. The K/DST network consisted of two fully connected hidden layers with 16 and 8 nodes respectively, followed by a single output layer. The Multi-Positional network consisted of two fully connected hidden layers with 32 and 16 neurons, respectively, followed by a dropout layer with a rate of 0.4 and a single linear output layer for predicting fantasy points.

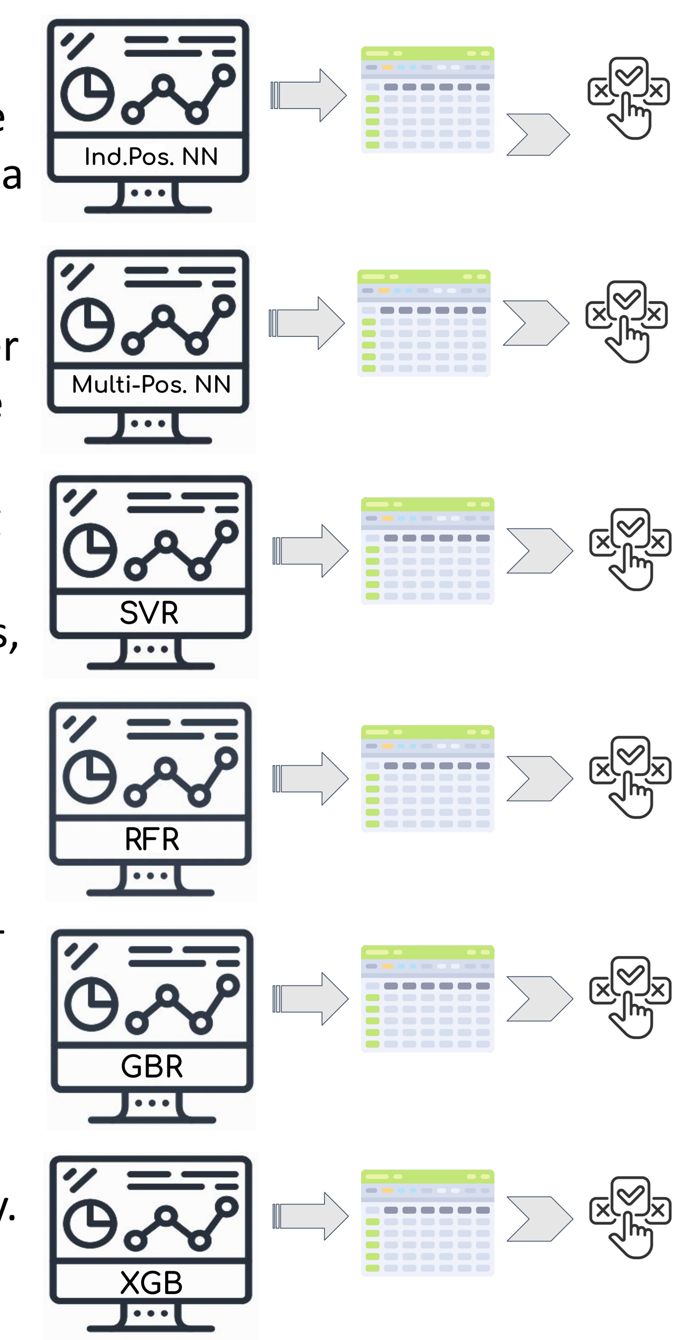
To evaluate the accuracy of the predictions from each model, I computed the R² value for every set of predictions each model made for each position, besides the Multi-Positional NN model, which just has one overall R² value.



Model Positional and Average R² Value

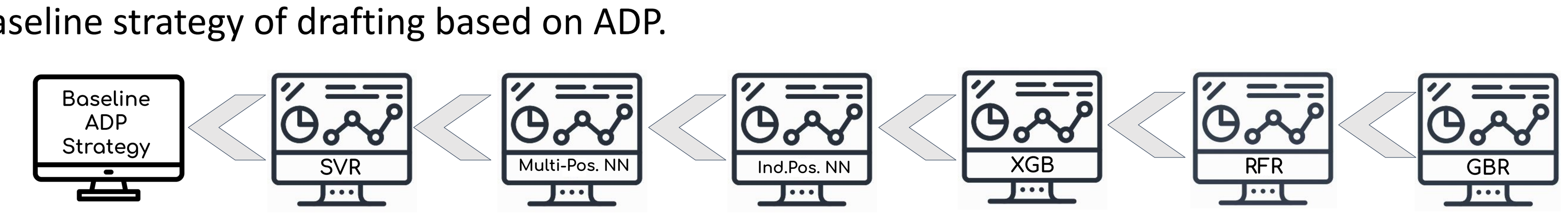
	QB	RB	WR	TE	K	DST	Avg Model R^2 Value
Ind. Pos NN	0.515	0.412	0.513	0.763	-0.181	-0.029	0.332
Multi-Pos. NN	-	-	-	-	-	-	0.619
SVR Model	0.058	0.426	0.324	0.514	0.794	0.275	0.399
RFR Model	0.674	0.916	0.845	0.948	0.913	0.762	0.801
GBR Model	0.605	0.929	0.857	0.966	0.840	0.965	0.860
XGB Model	0.624	0.934	0.854	0.940	0.787	0.875	0.836

Each model's predictions were used to compile a "Draft Board" sorted in descending order of the points the model predicted for the following season. In the draft simulations, the model was assigned a random pick in each simulation and the other 11 opponents in each simulation were drafting based on the baseline strategy.

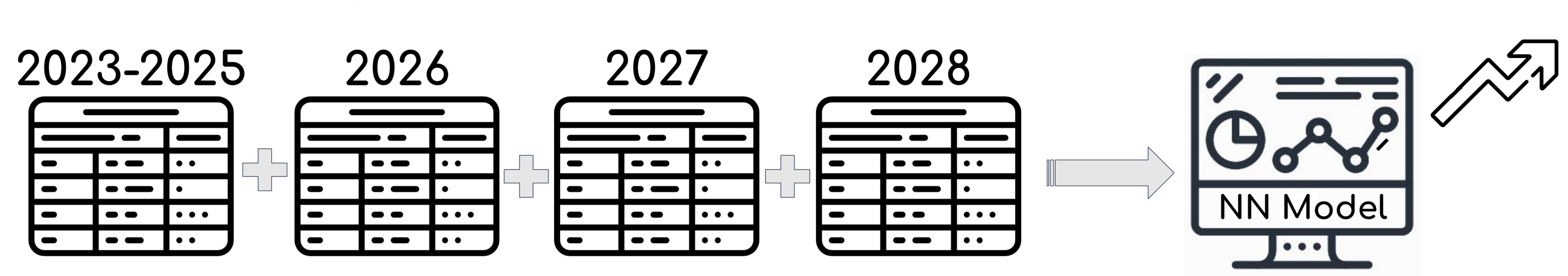


Discussion & Future Research

- All models, on average, outperformed the baseline strategy in 100 simulated drafts.
- Based on the performance in the draft simulations, the Gradient Boosting Regression Model was the top performing model out of the 6 that were analyzed. It was the model that most out-performed the baseline strategy of drafting based on ADP.

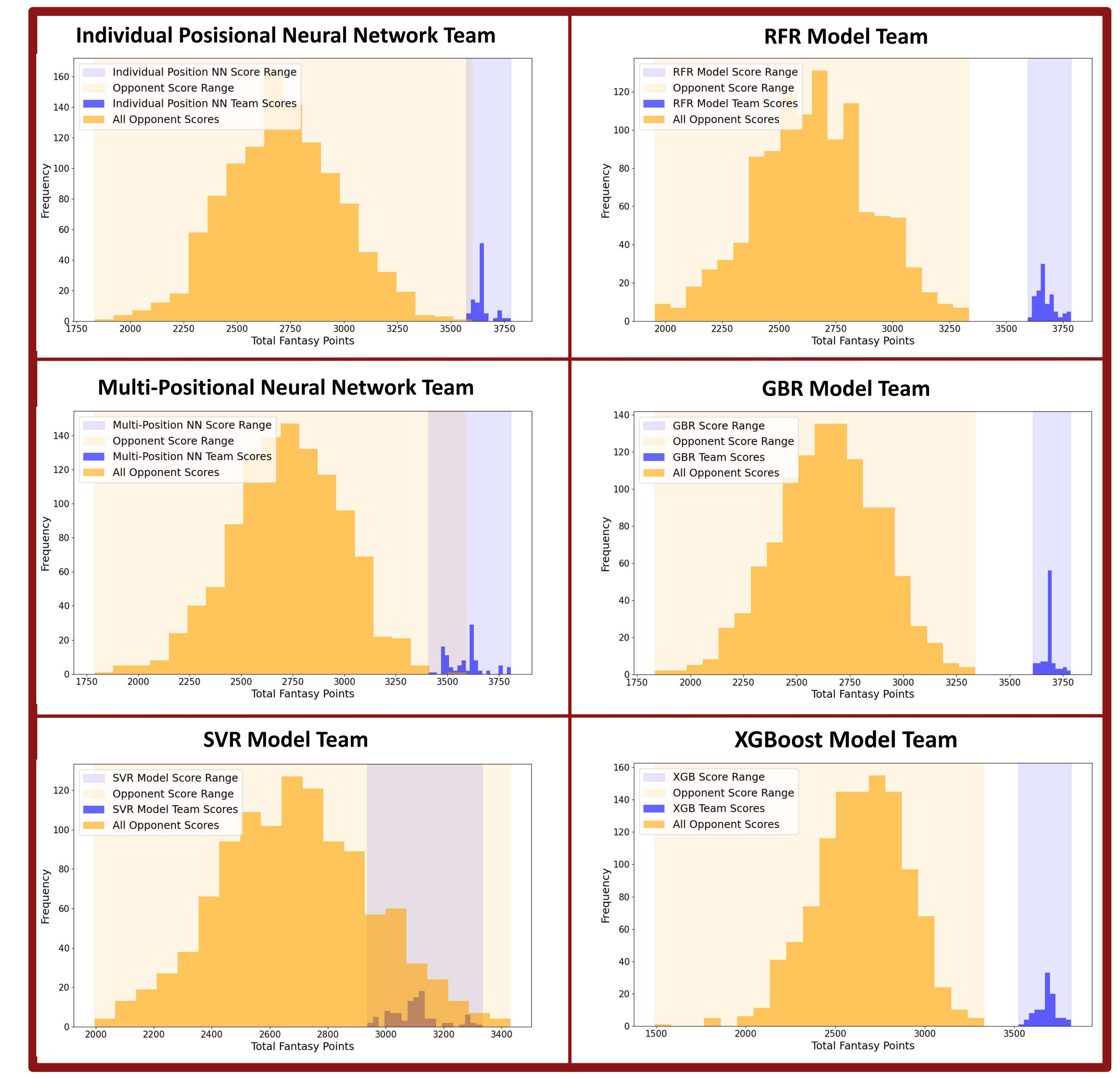


- Both neural networks out-performed the baseline strategy and the SVR model. The Individual Positional NN and the Multi-Positional NN likely underperformed due to limited data, and expanding the dataset in future seasons should allow them to learn deeper trends and potentially match or surpass Gradient Boosting.

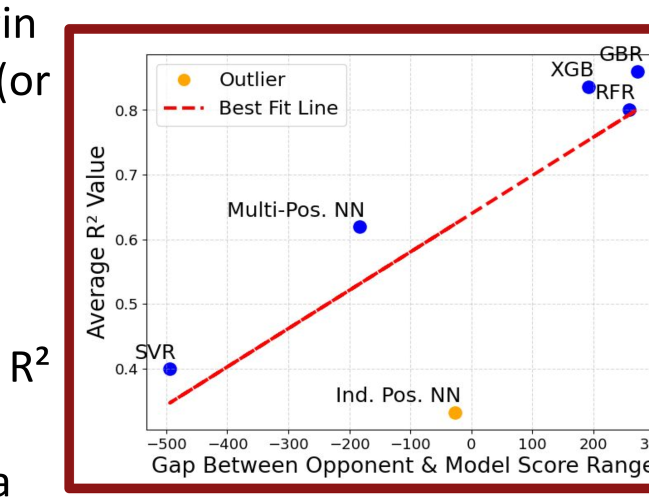


Results

Model Performance in Simulated Drafts



Plotting the average R² value against the margin at which the lowest scoring model team beat (or lost to) the highest scoring opponent team shows a linear relationship between the two metrics. The Individual Positional NN Model is an outlier in this relationship due to it's poor predictions for K and DST lowering its average R² value. However, those positions are far less impactful contributors to the overall score of a team.



Average Scores Over 100 Drafts

	Ind. Pos NN	Multi-Pos. NN	SVR	RFR	GBR	XGB
Model	3645.29	3587.41	3104.42	3673.49	3683.82	3682.05
Opponent	2715.67	2722.50	2687.39	2650.15	2648.47	2644.80
Average Win Margin ($ModelAVG - OppAVG$)	+929.62	+864.91	+417.03	+1023.34	+1035.35	+1037.24
Gap b/w Model and Opp ($lowestModel - highestOpp$)	-27.2	-183.7	-494.2	+259.2	+272.2	+192.2

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